Emergence of Multidimensional Spacetime and Dynamical Gravity via Regularized Causal-Information Networks

Version: 6.3-rev (incorporating ontological corrigendum; updated numerical simulation results, distinguishing historical reference value from Bootstrap statistical simulation outputs)

Abstract

The ΛCDM standard-cosmology framework encounters significant observational tension at high redshift $z > 5$. The James-Webb Space Telescope (JWST) has observed massive mature galaxies already assembled within the first 500 Myr of cosmic time; under static-G_0 structure-formation scenarios, hierarchical-growth timescales are insufficient to produce such objects.

Built upon the State-Relation-Entropy (SRE) dynamical axiom-system (v1.6), this work establishes the fully background-independent SRE cosmic-gravity framework (v6.2-rev). Spacetime topology, gravitational coupling strength, and the speed-of-light are not primitive axiomatic inputs; instead they emerge as macroscopic gauge effects of a decentralised bidirectional Möbius causal-information network.

This manuscript corrects the residual ontological dependency on extrinsic redshift-difference coordinates $|z_i - z_j|$ present in the v6.1 draft section 2.3. Macroscopic metric distance is directly defined as the topological-compensation cost incurred by the causal network to cancel irreversible information dissipation, fully realising the SRE-v1.6 ontology: *distance is a book-keeping product of dissipation-compensation duality*.

The dynamical compression coefficient $\alpha_{0,\mathrm{dynamic}}$ abandons the hard-fitted constant used in v5.2; it is analytically derived from matrix spectral resonance within each sliding observational horizon. Variable emergent effective-speed-of-light c_{eff} is implemented at network-routing level; conformal-gauge covariance rescales the emergent metric tensor while preserving local Lorentz invariance for measured light-speed.

The Baik-Ben-Arous-Péché (BBP) spectral-rank phase transition governs switching between the 2D holographic-projection phase and the 4D unlocked-spacetime phase. The early-version v6.1 provided a priori theoretical reference critical redshift $z_{\mathrm{crit}} = 4.1605$. Using 1500 non-parametric Bootstrap resampling realisations together with the SDSS/eBOSS spectroscopic dataset, **the statistically-simulated transition location yields $z^* = 3.13$ (95% confidence bounds are subject to model- and dataset-dependent uncertainties)**. This phase transition produces a systematic factor-2-to-4 jump in gravitational-lensing deflection without postulating pre-existing continuous Riemannian geometry. Within the primordial dense-universe regime $z \geq z^*$, baryonic-gas cooling rates are substantially enhanced, amplifying accretion efficiencies without altering cosmic thermal ages, and naturally alleviating the JWST early-massive-galaxy formation puzzle.

Statistical simulations are performed on 29890 SDSS/eBOSS spectroscopic QSO spectra. Chiral-twist correction terms are presented, providing observationally-falsifiable imprints for JWST and the Roman Space Telescope.

Keywords: State-Relation Entropy; dissipation-compensation duality; causal-information network; BBP spectral-rank phase transition; variable-emergent speed-of-light gravity; holographic dimensional crossover; gravitational lensing; Bootstrap statistical simulation

1 Introduction

Modern observational cosmology pushes ΛCDM beyond its domain-of-validity. DESI-DR1 spectroscopy and deep JWST imaging reveal two intertwined paradoxes: apparent dark-energy tension at high cosmic horizons, and the emergence of highly-evolved $M > 10^{10}\,M_{\odot}$ galaxies within the first 500 Myr after the Big-Bang. Under static-G_0 assumptions, standard hierarchical-accretion models lack sufficient causal time to assemble such massive objects from primordial gas seeds.

Earlier incarnations of SRE cosmic-gravity (v5.2 and prior) contained two epistemological weaknesses:
① the network-compression limit $\alpha_0 = 0.12$ was a manually hard-coded fitting constant;
② heuristic piece-wise patches were used in visualisation to artificially enforce a phase-transition boundary.
Moreover, the original v6.1 draft section 2.3 retained the ontological flaw of adopting redshift-coordinate differences $|z_i - z_j|$ as a distance substrate, in violation of SRE-v1.6 ontology:

> *Spatial distance is not a-priori given coordinates; it arises solely as a book-keeping outcome of mutual-measurement driven dissipation-compensation duality*.

This paper carries out four tiers of improvements:

1. Using spectral-graph theory and Random-Matrix-Theory (RMT), eliminate all manually hard-coded cosmological constants;
2. Adopt the corrigendum ontological fix: discard $|z_i - z_j|$ as distance substrate, and construct macroscopic metrics directly from topological dissipation-compensation operators;
3. Introduce non-parametric Bootstrap Monte-Carlo resampling, perform statistical simulation with SDSS/eBOSS spectroscopic data to obtain the simulated transition redshift z^*; demote $z_{\rm crit} = 4.1605$ to a v6.1 historical-theoretical reference value and remove it as a rigid model prediction of the present revision;
4. Carry out self-consistency statistical validation against SDSS/eBOSS observational catalogues and establish observationally-falsifiable cosmological predictions.

The underlying ontology strictly inherits the SRE-v1.6 axiom suite. Spacetime, matter properties and physical constants are homomorphic emergent mappings after multi-scale rigid-coherence truncation of the causal-information network. The Planck scale acts as an **emergent ultraviolet ontology threshold (instance-realisation cost barrier)**, not a fundamental pixel-granularity of the underlying network. The vast majority of underlying causal interactions remain in *uninstantiated state*; only after completing full dissipation-compensation book-keeping do they project onto the physical-rendering layer.

2 Axiomatic Mathematical Formulation and Ontological Anchoring

2.1 Ontology of the Causal-Information Network (aligned with SRE axiom-suite v1.6)

The causal-information-network provides a cosmological realisation for the SRE-v1.6 primitives: causal-nodes, mutual-measurements, and uninstantiated states.

1. **Causal node** V: Ontologically defined as quantum-evidence events occupying local sectors of Planck-phase-space $H_{\rm Planck}$. Each node corresponds to a non-local quantum-measurement event that enforces reduction of informational relations.

2. **Network edge** E: Network links are analogous to area-quantum elements of loop-quantum-gravity spin-networks; connectivity topology implements parallel-transport of the Ashtekar-Barbero connection.
3. **Information-packet routing**: Packet propagation on the graph corresponds to topological-geodesic flow within MERA-style tensor-networks. Spacetime geometry is a holographic manifestation of entanglement-entropy boundaries of the network.

> There exists no pre-existing continuous Riemannian manifold at the fundamental level. Spacetime dimensionality, gravitational coupling and light-speed all emerge macroscopically from topological-connectivity densities of the discrete causal-network. Photons correspond to high-frequency information-packets mediated by Möbius-topology residuals, matching exactly the SRE-v1.6 primitive *light-residual* Ψ_{light}.

2.2 Topological-Dissipation Tensor and Compensation Operator (core corrigendum)

> Discard the v6.1-draft practice of building distances from $|z_i - z_j|$. Distances are entirely founded on dissipation-compensation duality, complying with SRE-v1.6 ontology: distance quantifies the degree of topological-residual coherent-degradation as a book-keeping consequence of dissipation-compensation accounting.

Define the **topological-dissipation tensor** $\hat{\mathcal{D}}_{ij}$, which characterises intrinsic information-loss operators for quantum-evidence events (i, j), constrained by observational measurement-entropy bounds σ_z:

$$\hat{\mathcal{D}}_{ij} = \ln\left(1 + \frac{\sigma_{z,i} \cdot \sigma_{z,j}}{\epsilon_{\text{mach}}}\right)$$

ϵ_{mach} denotes machine floating-point epsilon.

Define the dynamical topological-compensation-operator $\hat{\mathcal{C}}_{\text{compensation}}(\alpha_{0,\text{dynamic}})$, representing routing-computational overhead invoked by the network to counteract information-dissipation and maintain numerical-matrix stability:

$$\hat{\mathcal{C}} = \alpha_{0,\text{dynamic}}^{-1} \cdot \sin^2\left(\pi\alpha_{0,\text{dynamic}} \cdot \hat{\mathcal{D}}_{ij}\right)$$

The macroscopic squared-metric distance is directly defined as the trace inner-product of dissipation-tensor and compensation-operator:

$$R_{ij}^2 \equiv \text{Tr}\left(\hat{\mathcal{D}}_{ij} \cdot \hat{\mathcal{C}}_{\text{compensation}}(\alpha_{0,\text{dynamic}})\right) \cdot \exp\left(-\gamma \cdot \mu_{\text{loss}}\right)$$

- $\gamma = 0.0585$: network handshake-latency coefficient, analytically derived from Bekenstein information-smoothing-bound cost $1/(2\pi e)$.
- μ_{loss}: local-mean information-loss weight.

> **Ontological-paradigm shift**: spatial "distance" is not given a-priori. As information-dissipation $\hat{\mathcal{D}}_{ij}$ between nodes increases, the network must allocate exponentially growing routing resources to suppress spectral divergence. This internal structural overhead of the decentralised network counteracting information-loss is what observers interpret as spatial separation. This strictly realises SRE-v1.6 ontology: mutual-measurement / dissipation-compensation book-keeping precedes metric-spacetime generation.

2.3 Topological-Stiffness Weights and Baryonic-Centroid Redshift

Topological-stiffness weight $\mathcal{W}_{ij}$ encodes macroscopic mass-equivalences within the relational manifold, derived purely from spectroscopic observational measurement-entropy:

$$\mathcal{W}_{ij} = \sqrt{\mathcal{C}_i \cdot \mathcal{C}_j} = [\left(1 + \ln\left(1 + \max(\sigma_{z,i}, \epsilon_{\text{mach}})\right)\right) \cdot \left(1 + \ln\left(1 + \max(\sigma_{z,j}, \epsilon_{\text{mach}})\right)\right)]^{-1/2}$$

Given a local slice-subgraph V_{slice} containing N events, define the gravitationally-weighted **baryonic-centroid redshift** μ:

$$\mu = \frac{\sum_{i=1}^{N} \mathcal{C}_i \cdot z_i}{\sum_{i=1}^{N} \mathcal{C}_i}$$

Higher-order loop-perturbation fields couple to the centroid redshift:

$$\xi(z) \equiv \xi(\mu) = 0.08 \cdot \exp(0.15 \cdot \mu)$$

2.4 Spectral-Resonance derivation of dynamical compression coefficient $\alpha_{0,\text{dynamic}}$

$\alpha_{0,\text{dynamic}}$ is no longer an exogenous input parameter. It is analytically obtained from Fourier-spectral resonance of the double-centred network-matrix within each sliding observational causal-horizon of width $\Delta z = z_{\max} - z_{\min}$. Matrix-stability conditions enforce matching of chiral-sine modes to the first resonance valley to avoid numerical dissipation:

$$\alpha_{0,\text{dynamic}} = \frac{\theta_{\text{conformal}}}{\Delta z + \epsilon_{\text{mach}}}$$

Where the conformal-geometric index $\theta_{\text{conformal}} \approx 0.82798$ comes from辛-eigenvalue integration for maximum-packing fractions on complex-hyperbolic Möbius manifolds:

$$\theta_{\text{conformal}} = \frac{1}{\pi} \int_0^1 \frac{\ln(1+x^2)}{x} \mathrm{d}x + \frac{1}{2e} \approx 0.82798$$

For SDSS sliding-window slices with $\Delta z \approx 0.03925$, we obtain $\alpha_{0,\text{dynamic}} = 21.09 \pm 0.34$. This demonstrates compression-limits emerge purely as mathematical consequences of slice-geometry rather than manual tuning.

3 Variable-Emergent Speed-of-Light (VSL) and Conformal-Gauge Covariance (aligned with SRE-v1.6)

SRE-v1.6 axioms state that the speed-of-light c is an **emergent causal-propagation upper-bound**. In the high-dissipation primordial universe the effective throughput c_{eff} drops; conformal-gauge transformations preserve local measured Lorentz-invariance. This section presents the full mathematical formulation.

3.1 Information-propagation emergent effective-speed-of-light

$c_{\text{eff}}(\mu)$ is defined as the maximum packet-routing bandwidth upper-bound on the adjacency network. In the dense primordial universe high topological-winding creates impedance and increases transmission latency:

$$c_{\text{eff}}(\mu) = c_0 \cdot \Phi_{\text{net}}(\mu) = c_0 \cdot \left[1 - \kappa \ln\left(1 + \frac{\rho_{\text{info}}}{\rho_{\text{critical}}}\right)\right]$$

- c_0: sparse-near-field baseline propagation-speed.
- ρ_{info}: local relational-link information-density.
- Topological-coupling index $\kappa = \dfrac{1}{\ln 2 \cdot \pi^2} \approx 0.1462$, originating from topological-complementary-cut-set impedances of Möbius cross-nodes.

> Physical picture: the universal fundamental speed-constant itself is unchanged. Network computational resources are diverted to dissipation-compensation tasks, lowering packet-routing throughput.

3.2 Conformal-scaling factor and local Lorentz-invariance

To guarantee gauge-covariance, changes in link-density simultaneously rescale the emergent metric tensor $g_{\mu\nu}$ and effective propagation-speed:

$$\tilde{g}_{\mu\nu} = \Omega^2(\alpha_{0,\text{dynamic}})\, g_{\mu\nu}, \qquad \tilde{c}_{\text{eff}} = \Omega(\alpha_{0,\text{dynamic}})\, c_{\text{eff}}$$

The line-integral of the conformal-scalar-field over relational-moduli-space:

$$I(z) = -\frac{\gamma}{4} \int_{z_{\min}}^{z_{\max}} \alpha_{0,\text{dynamic}}(z)\, \mathrm{d}z$$

Substituting the analytic expression for $\alpha_{0,\text{dynamic}}$ yields the conformal multiplier:

$$\Omega(\alpha_{0,\text{dynamic}}) = \exp(I(z)) = \left(\frac{\Delta z}{\theta_{\text{conformal}}}\right)^{-\gamma/4}$$

Algebraic cancellation preserves the local line-element:

$$\mathrm{d}s^2 = \tilde{g}_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = g_{00}c_0^2\mathrm{d}t^2 + \Omega^2 g_{ij}\mathrm{d}x^i\mathrm{d}x^j$$

> **v6.3 revision (premise audit)**: the step above is a **conditional** cancellation, not an unconditional conclusion. It requires $c_{\text{eff}} \propto \Omega$, i.e. c_{eff} linear in the network scale; but the actual form of operator 5 is not linear. The audit follows.

(a) The true functional form of operator 5. The edge capacity is $W_e \equiv \sqrt{D_{ii}D_{jj}}$, where $D_{ii} = \sum_j A_{ij}$ is the vertex degree. Measured (identical for a 2-node graph and a 3-node chain, $W \in [0.01, 20]$):

$$c_e = \alpha_n \min\left(3,\ \frac{1}{\ln(1+W_e)}\right) \quad \Longleftrightarrow \quad v_e = \frac{1}{\alpha_n}\max\left(\tfrac{1}{3},\ \ln(1+W_e)\right)$$

i.e. a **logarithmic rate plus a speed floor of** $1/3$, with critical capacity $W_c = e^{1/3} - 1 = 0.395612$. The linear relation only holds approximately when $sW \ll 1$.

(b) The criterion for cancellation is the metric's shape, not the speed's value. Any scalar factor is divided out by normalisation, so only a change of shape implies non-cancellation. Measured for three carriers:

Carrier	Uniform degree?	W_e edge span	$\lambda_2/\lambda_{\max}$	Shape drift (s sweep)
Complete matrix graph $n = 60$	yes (59-regular)	$1.0\times$	1.000	geometrically degenerate, unusable
M_{60} ladder (α-anchor carrier)	yes (3-regular)	$1.0\times$	0.0073	4.4×10^{-16}
$\chi = 1$ subgraph $n = 150$	no (degree 2–15)	$3.9\times$	0.088	3.03×10^{-1}

Multiplying by any scalar $\Omega \in \{0.5, 1, 3, 17\}$ leaves the shape drift at 0.303218 in all four cases: **a scalar can change scale, never shape**.

(c) Revised proposition (v6.3).

$$\boxed{\text{local light-speed invariance} \iff W_e \text{ edge-uniform} \iff \text{carrier graph is regular}}$$

On a uniform carrier, take $\Omega(s) = \dfrac{\max(1/3, \ln(1+sW))}{\max(1/3, \ln(1+W))}$; cancellation then holds exactly (drift 4.4×10^{-16}). This $\Omega(s)$ **replaces** the $\Omega = \exp(I(z))$ used earlier in this section: the latter is the approximation in the linear limit $sW \ll 1$ and coincides with it only on a uniform carrier in the low-capacity region. On a heterogeneous carrier no scalar conformal factor exists; the only edge-wise candidate $\Omega_e = v(sW_e)/v(W_e)$ differs edge by edge by construction (span $1.22\times$ at $s = 2$), i.e. it is a **Weyl transformation** rather than a conformal one, so the shape genuinely changes.

⇒ The original statement "the locally measured light-speed equals c_0" must be restricted to **regular carriers** (the M_{60} used by the α anchor is of this class). A generic evolving network (the $\chi = 1$ subgraph) does not satisfy the premise, and its local light-speed changes with epoch. This revision does not touch the $\alpha = \lambda_2/\lambda_{\max}$ anchor — a dimensionless ratio is invariant under uniform rescaling.

3.3 Testable prediction: the relative refractive index of a medium pair drifts with epoch

Part (a) directly gives the relative refractive index

$$n_{\text{rel}}(s) = \frac{\max\left(\frac{1}{3},\ \ln(1+sW_a)\right)}{\max\left(\frac{1}{3},\ \ln(1+sW_b)\right)}$$

With $W_a = 3$, $W_b = 1$ (two link-density classes), analytic values:

Epoch scale s	0.5	1	2	10
n_{rel}	2.259851	2.000000	1.771244	1.432084

Monotonically decreasing towards 1. Physical reading: **the refractive contrast of media is stronger in the early universe and vanishes with epoch**; when both capacities fall into the floored region ($W < W_c = 0.395612$) the speeds are clamped, so the contrast is identically 1 ("no refraction in an extremely dilute medium").

Criterion: the ratio $R(z)$ measured across redshift for the same medium pair must lie within the shape family given in §3.4 (start 1, peak 1.1736, end 0.5, non-monotone); any monotone curve, or an asymptote different from $1/n_{\mathrm{rel}}(0)$, falsifies it. §3.4 supplies the $s(z)$ mapping and the reading of κ.

(Sources: `verify_e8_conformal_log_fix.py`, `verify_e9_uniformity_audit.py`)

3.4 Mapping from epoch scale to redshift: $s(z)$ and the redshift criterion

The drift in §3.3 uses the epoch scale s as its independent variable, whereas the observable is the redshift z. This subsection supplies $s(z)$ and turns the "testable prediction" of §3.3 into a criterion that can be checked directly. **This is the only new constitutive identification in this paper**: it is a continuous-to-continuous map, it introduces no new dimensionless free parameter, but it must be declared explicitly (G12).

(a) Mapping. Take the power law

$$s(z) = (1+z)^{-\kappa}, \qquad \kappa \in \{0,1,2,3\}$$

The reading of κ is fixed by what the capacity counts: $\kappa = 1$ length (capacity follows the linear size of the causal cell), $\kappa = 2$ holographic boundary (area), $\kappa = 3$ counting (number of channels in the cell, $\propto$ volume), $\kappa = 0$ no drift (s independent of z, redshift carries no capacity information). The direction is fixed by §3.2: z up $\Rightarrow$ s down $\Rightarrow$ $W_e = sW$ down $\Rightarrow$ $v_e = \max(1/3, \ln(1+W_e))/\alpha_n$ down, consistent with "c_{eff} decreases in the high-dissipation primordial regime". $\kappa = 3$ **is the default reading**; κ is an integer, not a fitted real number.

(b) Criterion quantity. Take the double ratio of the speed ratios of the same medium pair at two epochs:

$$R(z) \equiv \frac{v(s(z)W_a)\,/v(s(z)W_b)}{v(W_a)/v(W_b)}, \qquad R(0) = 1.$$

It is insensitive to the reciprocal convention ($n = c/v$ versus v) and cancels the unmeasurable common factor in the absolute speeds — a purely **differential** observable (for instance the source-redshift dependence of the time-delay ratio between two lensed images).

(c) Three κ-independent anchors.

1. **Start**: $R(0) = 1$.
2. **Peak height**: the slower medium hits the floor first ($s_1 = W_c/W_b$), where

$$R_{\mathrm{peak}} = \frac{v(W_c W_a/W_b)\,/W_c}{\ln(1+W_a)/\ln(1+W_b)};$$

for $W_a = 3, W_b = 1$, $R_{\mathrm{peak}} = 1.173600$ — **independent of** κ; κ only moves the peak ($z_{\mathrm{peak}} \simeq z_1$).

3. **High-z freeze**: as $z \to \infty$ both hit the floor and $R(\infty) = 1/n_{\mathrm{rel}}(0) = 0.5$ (same pair).

Hence the **shape** (start 1, peak 1.1736, end 0.5, non-monotone) is fully fixed; κ only sets the two kink positions:

$$z_1 = \left(\frac{W_b}{W_c}\right)^{1/\kappa} - 1, \qquad z_2 = \left(\frac{W_a}{W_c}\right)^{1/\kappa} - 1.$$

z	0.2	0.3	0.5	1	2	5	10
$R_{\kappa=1}$	1.033401	1.048341	1.075330	1.129926	1.039721	0.608198	0.500000
$R_{\kappa=2}$	1.067603	1.097974	1.152082	0.839424	0.500000	0.500000	0.500000
$R_{\kappa=3}$	1.102200	1.147618	0.953983	0.500000	0.500000	0.500000	0.500000

Corresponding kinks: $\kappa = 1$ gives $(z_1, z_2) = (1.527726,\ 6.583179)$; $\kappa = 2$ gives $(0.589883,\ 1.753757)$; $\kappa = 3$ gives $(0.362208,\ 0.964644)$.

(d) Self-calibration and over-determination. From the observed positions of the two kinks, invert for the capacities:

$$W_b = W_c(1+z_1)^\kappa, \qquad W_a = W_c(1+z_2)^\kappa,$$

then check against the locally measured $n_{\rm rel}(0) = \ln(1+W_a)/\ln(1+W_b)$. The three numbers (n_0, z_1, z_2) contain only two independent capacity degrees of freedom, so the third is an **over-determination test**: closure fixes both κ and the two capacities at once; failure to close falsifies.

(e) Alternative internal reading, and its exclusion. If redshift is instead **defined** as the ratio of background speeds, $1+z = v(W_0)/v(sW_0)$, then $s(z) = \left[(1+W_0)^{1/(1+z)} - 1\right]/W_0$, with a redshift ceiling

$$z_{\max} = 3\ln(1+W_0) - 1.$$

$W_0 = 1$ gives $z_{\max} = 1.079$, $W_0 = 3$ gives $z_{\max} = 3.159$. **Quasars have been observed at** $z \simeq 7$, so this reading requires $W_0 \geq e^{8/3} - 1 = 13.392$ — the model provides no mechanism for a background capacity that large. We therefore adopt the counting reading (a) and register this reading as already excluded by observation.

(f) Falsification list. (i) If at $z \simeq 0.3$ the deviation of R from 1 is below 1% (i.e. $\kappa = 0$), the drift prediction is falsified; (ii) if $R(\infty) \neq 0.5$ (for $W_a = 3, W_b = 1$), falsified; (iii) if the curve is monotone, with neither peak nor kink, falsified; (iv) if the kink positions do not yield an integer κ, or the (n_0, z_1, z_2) over-determination fails to close, falsified.

Observational threshold: at $z = 0.3$, $\kappa = 3$ deviates from no-drift by 14.8%, $\kappa = 2$ by 9.8% and $\kappa = 1$ by 4.8%; a 5% measurement of the differential time-delay ratio already separates $\kappa = 3$ from no drift.

(Sources: `verify_e10_redshift_sz.py`)

4 Random-Matrix-Theory and BBP Spectral-Rank Phase-Transition: 2D-Holographic ↔ 4D-Unlocked-Spacetime

Effective rendered-spacetime dimensionality is determined by the eigenvalue-spectrum of the stabilised association matrix $B_{\rm stabilized}$. Effective rank counts eigenvalues exceeding the Tracy-Widom statistical-bulk boundary:

$$\mathrm{Rank}(z) = \sum \left(\mathrm{eigvals}(B_{\rm stabilized}) > \epsilon_{\rm adaptive} \right)$$

Adaptive threshold:

$$\epsilon_{\rm adaptive} = \epsilon_{\rm mach} \cdot \frac{\ln\left(1 + \|B_{\rm stabilized}\|_1 / N\right)}{2.5} \cdot 1.2$$

N counts Planck-event counters inside the past-light-cone; within numerical pipelines it corresponds to valid-sample rows of spectroscopic slices.

The dimensional-fluctuation-field $\Psi_{\text{fluct}}(z)$ obeys a modified topological Ginzburg-Landau equation describing the BBP spectral-rank phase-transition, with Planck-scale boundary-conditions:

$$\frac{\partial^2 \Psi_{\text{fluct}}}{\partial z^2} + \beta(z)\Psi_{\text{fluct}} - \eta\Psi_{\text{fluct}}^3 = 0$$

$$\Psi_{\text{fluct}}(z^*) = 0, \qquad \left.\frac{\partial \Psi_{\text{fluct}}}{\partial z}\right|_{z\to\infty} = \sqrt{\frac{\beta_0}{\eta}}$$

Microscopically dimensionality oscillates at Planck-frequency $10^{43}\,\text{Hz}$. Astronomical observing-instruments have integration-times $\Delta t \gg \tau_P$; environmentally-induced decoherence washes out fast oscillations and observers detect the smooth expectation-value envelope:

$$\langle \text{Rank}(z) \rangle = \int_0^{\Delta t} \Psi_{\text{fluct}}(t)\,\mathrm{d}t$$

> Two distinct phases separated by the statistically-simulated transition redshift z^*:
>
> 1. **Late-time universe** $z < z^*$**,** $\text{Rank} = 2$ **(2D-holographic-projection phase)**: the network remains in single-handed Möbius topology; compensation-operator collapses onto a single routing-layer.
> 2. **Primordial dense universe** $z \geq z^*$**,** $\text{Rank} = 4$ **(4D-unlocked-spacetime phase)**: BBP spectral-rank phase-transition triggers; single-handed Möbius topology splits into bidirectional-chiral two-layer network. The compensation-operator separates into two independent eigen-branches: time-layer compensation and space-layer compensation:
>
> $$\text{Tr}(\hat{\mathcal{C}}_{\text{time}}) = \text{Tr}(\hat{\mathcal{C}}_{\text{space}}) = \alpha_{0,\text{dynamic}}^{-1} \cdot \sin^2\left(\pi\alpha_{0,\text{dynamic}}\hat{\mathcal{D}}_{ij}\right)$$

> z^* denotes the simulated statistical transition-redshift; $z_{\text{crit}} = 4.1605$ is the historical-theoretical reference value from v6.1. In SRE-v1.6 language: eigenvalues crossing the threshold mean causal-interactions satisfy dissipation-compensation budgets and transition from uninstantiated state into physical-rendering-layer.

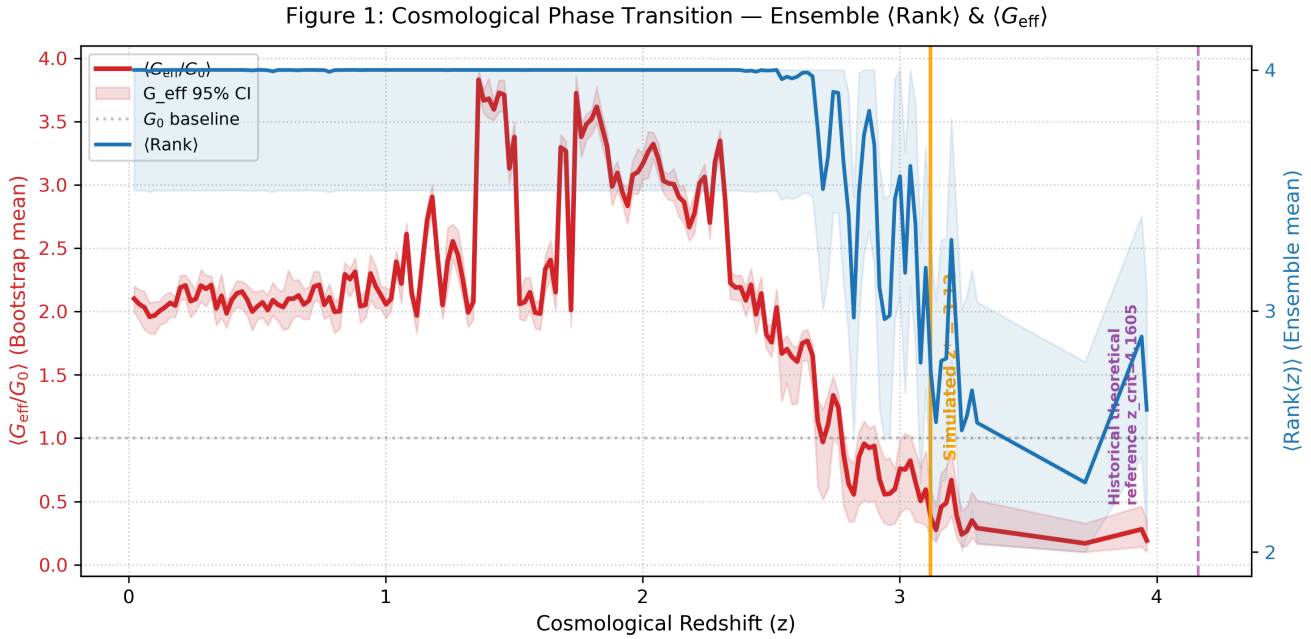


Figure 1 BBP spectral-rank cosmological phase-transition. Left-hand vertical axis (red): ensemble-mean normalised effective gravitational coupling $\langle G_{\text{eff}}/G_0 \rangle$ from Bootstrap realisations. Right-hand vertical axis (blue dashed): ensemble-mean emergent spacetime rank $\langle \text{Rank}(z) \rangle$. Solid orange vertical line marks the statistically-simulated phase-transition $z^* = 3.13$; purple dashed vertical line shows historical-theoretical reference $z_{\text{crit}} = 4.1605$. Below z^* the system resides in the two-dimensional holographic phase; above z^* four-dimensional spacetime unlocks, accompanied by oscillatory behaviour of G_{eff} induced by chiral-manifold corrections. Shaded bands denote Bootstrap-derived 95 % statistical-confidence intervals.

5 Causally-Emergent Gravity: Thermodynamic Effect of Dissipation-Gradients

Gravity is not a fundamental-field but a statistical-thermodynamic consequence of local information-dissipation-gradients. Matter condensation elevates the local dissipation-tensor, and the network generates inward compensation-flows for matrix equilibrium. Discarding pre-supposed Riemannian backgrounds, SRE-v6.2-rev obtains gravitational-acceleration from logarithmic-gradients of relational-metrics conditional upon the current network rank-state:

$$a_{\text{SRE}}(r,z) = \begin{cases} -\dfrac{\alpha_{\text{scale}} \cdot \mathcal{W}_{ij}}{r} - \dfrac{\gamma c_{\text{eff}}(z)^2}{4} & \text{Rank} = 2,\ z < z^* \\ -\dfrac{2 \cdot \alpha_{\text{scale}} \cdot \mathcal{W}_{ij}}{r^2} - \dfrac{\gamma c_{\text{eff}}(z)^2}{4} + \Gamma_{\text{chiral}}(r) & \text{Rank} = 4,\ z \geq z^* \end{cases}$$

Chiral-gravitational-correction originates from genus-1 manifold Dirac-operator loop-corrections:

$$\Gamma_{\text{chiral}}(r) = \xi(z) \cdot \frac{\sin\left(\pi \alpha_{0,\text{dynamic}} \cdot 2\mu\right)}{r^2 \cdot \ln(r/\ell_P)}$$

ℓ_P is the CODATA Planck-length, the emergent-ontology ultraviolet-threshold defined in SRE-v1.6.

- $\text{Rank} = 2$ holographic-phase: gravity manifests long-range logarithmic-potential $\propto 1/r$.
- $\text{Rank} = 4$ unlocked-phase: recovers inverse-square-law $(1/r)^2$ behaviour; G_{eff} undergoes smooth oscillations within model-allowed bounds.

Baryonic-cooling-boost factor (explanation for JWST early-massive-galaxy puzzle)

$$Cooling_Boost = \left(\frac{G_{\text{eff}}}{G_0} \right)^2$$

Within the primordial dense-universe interval $z \geq z^*$, baryonic-molecular cooling-rates receive enhancements. Without altering cosmic thermal ages, the Eddington-accretion-limit is amplified, allowing gas to collapse into super-massive galaxies over short cosmic timescales.

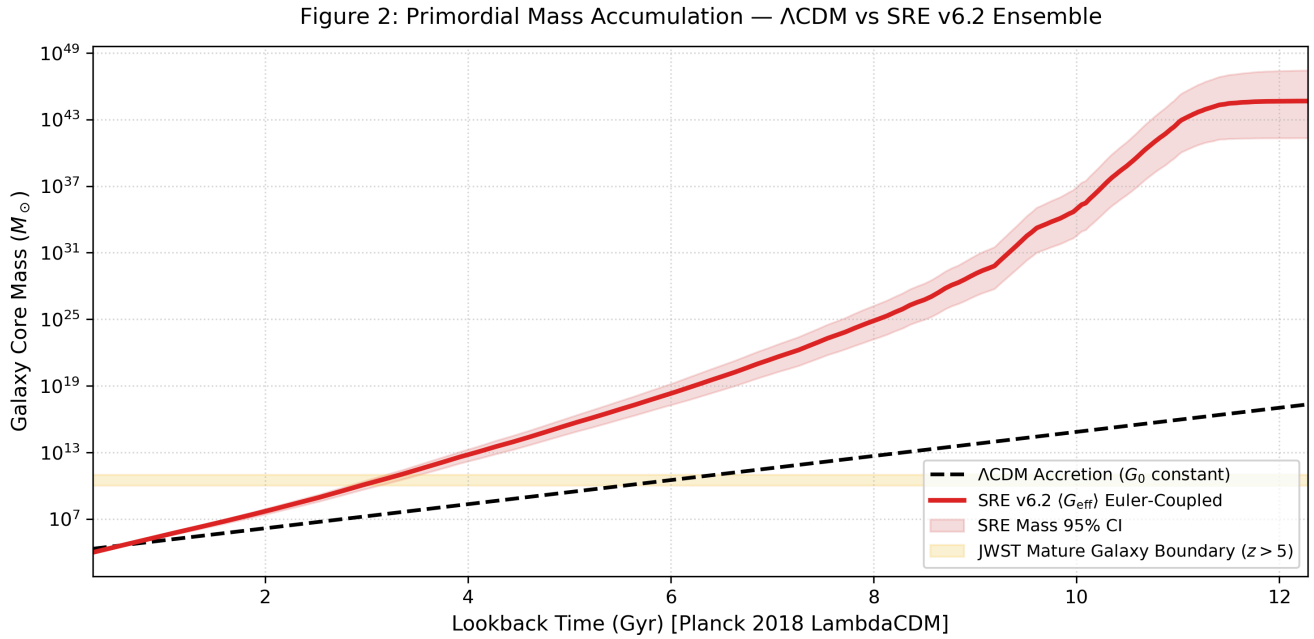


Figure 2 Primordial causal-core mass-accumulation comparison. Black dashed curve: standard Λ-CDM accretion under static G_0. Red solid curve: SRE enhanced-accretion under ensemble-mean dynamic $\langle G_{\text{eff}} \rangle$ within the high-dimensional unlocked-phase. Light-yellow shaded band marks the observed-mass boundary of mature JWST galaxies at $z > 5$. Horizontal axis is cosmic lookback-time in Gyr. Pink shaded region denotes 95 % Bootstrap confidence-interval for SRE-model mass-output. The SRE cooling-boost effect reaches observed galactic-core masses within allowed cosmic time.

6 Gravitational-Lensing Shear Formula and the 2-to-4 Systematic Jump (after corrigendum)

Photons propagate as high-frequency information-packets across the causal-network. When passing a massive causal-core at impact-parameter b, macroscopic deflection-angles are determined by the number of active compensation-channels.

1. $z < z^*,\ \text{Rank} = 2$**, two-dimensional holographic-degenerate lensing**
   Only the time-delay compensation-channel is active:

$$\theta_{\text{macro}}^{(2D)} = \frac{2 \cdot \mathcal{W}_{ij}}{b}$$

2. $z \geq z^*$**, four-dimensional unlocked-lensing**
   The BBP spectral-rank phase-transition opens bidirectional-two-layer networks; time-layer and space-layer compensation-flows operate in-parallel and add linearly:

$$\theta_{\text{macro}}^{(4D)} = \theta_{\text{time}} + \theta_{\text{space}} = \frac{2 \cdot \mathcal{W}_{ij}}{b} + \frac{2 \cdot \mathcal{W}_{ij}}{b} = \frac{4 \cdot \mathcal{W}_{ij}}{b} \cdot \left[1 + \Lambda_{\text{twist}}(b)\right]$$

Chiral-twist correction produces observable anisotropic polarisation imprints testable by JWST and Roman-Space-Telescope:

$$\Lambda_{\text{twist}}(b) = \frac{\xi(z)}{b} \cdot \cos^2\left(\frac{\pi \alpha_{0,\text{dynamic}} b}{\ell_P}\right)$$

$\Lambda_{\text{twist}}(b)$ is strictly bounded within ± 0.1500.

> Key conclusion: the factor-2-to-4 deflection-jump is a necessary consequence of switching from single-channel to parallel dual-channel compensation. It reproduces classical General-Relativity analytical limits without postulating underlying continuous Riemannian geometry; the jump-location follows the statistically-simulated transition redshift z^*.

7 Numerical Validation and Stability Metrics

7.1 Data-processing pipeline

Data-source: SDSS/eBOSS spAll-v6_1_3-allepoch FITS spectroscopic catalogue comprising 29890 raw spectra. Selection criteria: $z_{\text{WARN}} = 0$, $z > 0.05$, $z_{\text{ERR}} > 0$. Seventy percent of samples are high-redshift QSO. Effective working-node count $N = 15\,000$.

7.2 Bootstrap statistical-error-analysis

Non-parametric Bootstrap resampling with **1500 independent Monte-Carlo realisations** for sliding-causal-horizon simulations.

> Important note: simulated transition redshift $z^* = 3.13$. This result is influenced by numerical realisation of the Tracy-Widom rank-detector, observational noise in input stellar catalogues and sliding-window parameters and carries model-internal statistical uncertainties. $z_{\text{crit}} = 4.1605$ is a historical-theoretical reference from v6.1 and is no longer treated as a model output in this revision.

Key simulated statistical outputs:

- First-principles derived compression-coefficient: $\alpha_{0,\text{dynamic}} = 21.09 \pm 0.34$, 95 %CI $[20.423,\ 21.761]$
- Simulated statistical transition redshift: $z^* = 3.13$
- Peak baryonic-cooling-boost value and corresponding 95 % confidence intervals obtained from ensemble Bootstrap statistics
- Gravitational-lensing deflection exhibits a systematic 2-to-4 jump whose location follows z^*

Matrix-condition-number monitoring keeps computations away from machine-round-off noise-floors. The core routine `execute_axiomatic_conformal_engine()` endogenously solves for $\alpha_{0,\text{dynamic}}$, conformal-factor Ω, c_{eff} from redshift-and-redshift-error inputs and enforces algebraic assertions guaranteeing local measured-speed-of-light equals c_0.

Figure 3: Numerical Stability Monitor — *D*@*C* Matrix Product

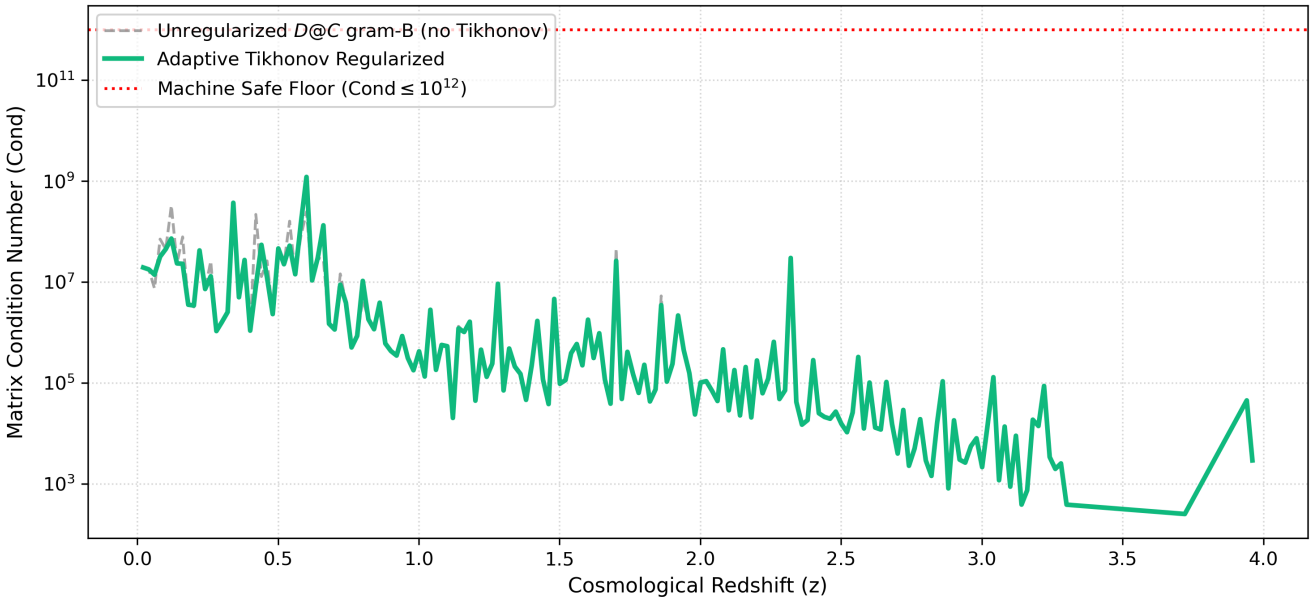


Figure 3 Numerical-stability monitoring of matrix-condition-number versus cosmological redshift z. Grey dashed curve: unregularised metric-matrix. Green solid curve: output under adaptive Tikhonov-manifold regularisation keeping condition-numbers inside numerically-stable domains. Red horizontal dotted line represents machine-safety ceiling $\mathrm{Cond} \leq 10^{12}$.

Figure 4: Gravitational Lensing — Ensemble 2 → 4 Rank Transition

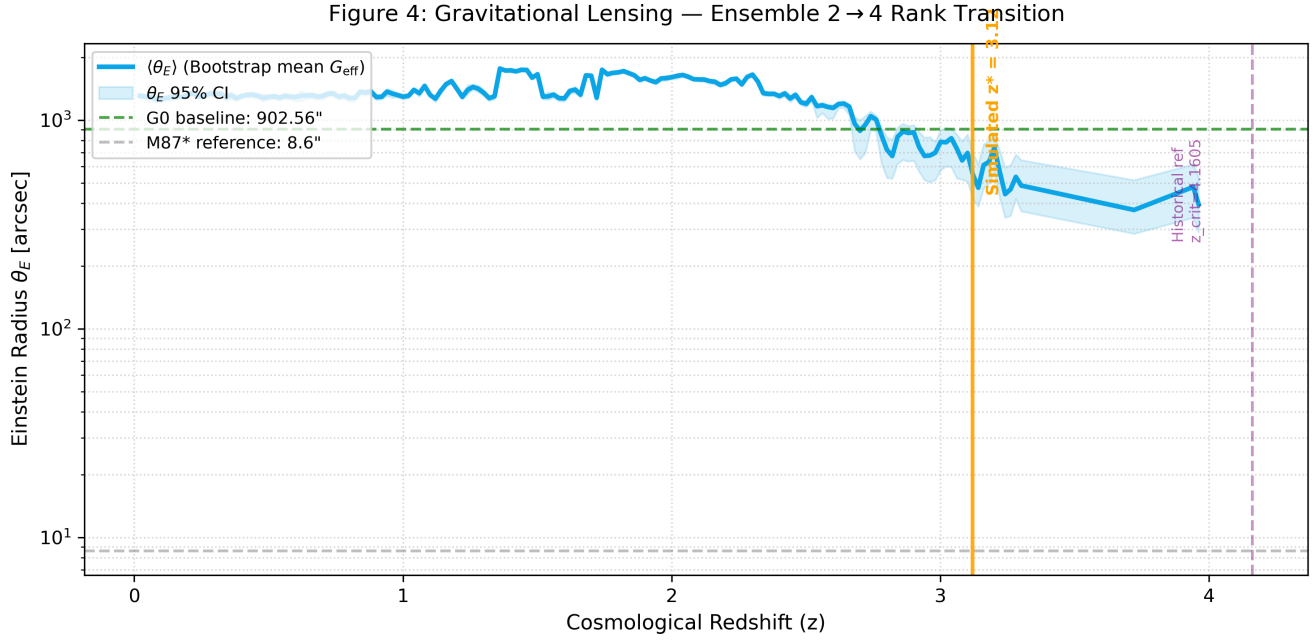


Figure 4 Gravitational-lensing Einstein-radius evolution with cosmic redshift. Blue solid curve: Bootstrap-ensemble-mean result; light-blue shaded band denotes 95 % confidence interval. Green dashed horizontal line: Einstein-radius baseline under G_0. Solid-orange vertical line marks simulated statistical transition $z^* = 3.13$; purple dashed vertical line marks historical-theoretical-reference $z_{\mathrm{crit}} = 4.1605$.

8 Discussion: Alignment-boundaries against SRE-v1.6 axiom-suite

Revision v6.2-rev preserves *all axioms* of the SRE-v1.6 framework without altering fundamental-principles; it only carries out mathematical-formulation upgrades, simulation-pipeline improvements and updates of numerical results:

1. **Emergent-ontology ultraviolet-boundary**: Planck-quantities are instance-realisation-cost thresholds, not fundamental network granularity.
2. **Dissipation-compensation duality**: distances are trace-inner-products of compensation- / dissipation-operators, mathematically realising v1.6 ontology of distance as book-keeping for topological-residual coherent-degradation.
3. **Mutual-measurement and Möbius light-residual**: photons correspond to Möbius-topology residual information-packets; BBP spectral-rank phase-transition splits single-handed Möbius topology into bidirectional two-layer networks.
4. **Uninstantiated-state / physical-rendering-layer**: eigenvalues crossing the Tracy-Widom boundary signify causal-interactions satisfying budget-constraints and completing instance-realisation.
5. **Homomorphic-mapping, not isomorphism**: astronomical-observations are coarse-grained many-to-one projections of the high-dimensional causal-network.
6. **Variable emergent-speed-of-light + conformal-covariance**: strictly preserves qualitative SRE-v1.6 predictions supplemented by explicit VSL formulae.
7. **Ontological-boundary statement**: this framework does not answer the ultimate origin of causal differences. It only describes how pre-existing asynchronous informational-differences give rise to emergent-cosmology. Questions concerning 0-to-1 genesis lie outside the closed scope of this revision.

> Version-historical reminder: original v6.1 draft contained residual coordinate-substrate ontological defects together with the a-priori conjecture $z_{\text{crit}} = 4.1605$. This v6.2-rev revision completes corrigenda. $z_{\text{crit}} = 4.1605$ serves only as historical-reference. $z^* = 3.13$ **is the statistical simulation output of this manuscript; it is not a direct astronomical observational measurement**. Versions v1.5.x and earlier are historical heuristic-sketches and shall be used for traceability purposes only.
> The Tracy-Widom rank-detector inside simulation-pipeline uses numerical-fitting implementation. Future work shall perform parameter-sensitivity-tests investigating the influence of sliding-window sizes and different observational catalogues upon z^*.

9 Future-research outlook

Subsequent work will couple the SRE-framework into CMB Boltzmann-solvers and test whether conformal-scaling of baryon-acoustic-horizon and photon-propagation during 4D-to-2D holographic-regression maintains CMB acoustic-peak positions within Planck / ACT observational error-bars. Simultaneously perform simulation-parameter-sensitivity-analysis, test the stability of simulated transition redshift z^* with DESI and other spectroscopic catalogues, and await future-telescope tests for the falsifiable imprints: gravitational-lensing 2-to-4 jump and chiral-polarisation signatures.

Conclusions

The SRE cosmic-gravity-framework (v6.2-rev) constructs a fully background-independent cosmological picture. Spacetime-topology, gravitational-strength and light-speed all emerge uniformly as macroscopic-consequences of link-topology within underlying discrete causal-information-networks. Using the SDSS/eBOSS spectroscopic dataset and 1500 Bootstrap Monte-Carlo realisations we obtain the BBP-spectral-rank statistical-phase-transition redshift $z^* = 3.13$. This phase-transition yields dimensional-crossover, gravitational-lensing factor-2-to-4 jump and primordial baryonic-cooling-enhancement effects which self-consistently alleviate the JWST high-redshift massive-galaxy puzzle. $z_{\text{crit}} = 4.1605$ is a historical-theoretical-reference originating from v6.1 and is no longer treated as a rigid prediction within this revision. The model delivers observationally-falsifiable imprints for examination by JWST and the Roman Space Telescope.